

2024

COMPUTER SCIENCE — HONOURS

Paper : SEC-3

(Mobile App Development)

Full Marks : 50

The figures in the margin indicate full marks.

*Candidates are required to give their answers in their own words
as far as practicable.*

Answer **question no. 1** and **any four** questions from the rest.

1. Answer **any five** questions : 2×5
- (a) What is a widget in Flutter?
 - (b) What is Dart and why it is used?
 - (c) What is Scaffold? Why it is used?
 - (d) What is stateless widget? Explain with appropriate illustrations.
 - (e) What is the purpose of hot reload and how different is it from hot restart?
 - (f) What is Android Studio? Highlight the key features.
 - (g) What is the purpose of final or const with respect to Dart programming?
 - (h) What is the use of safe area widgets? Explain with suitable illustrations.
2. (a) What is Android Virtual Device (AVD) and why is it used?
- (b) Can we use our own mobile phone as an alternative to AVD during app development? Justify your answer.
- (c) What settings need to be enabled on our mobile phone to use it instead of AVD?
- (d) Highlight two key features of Android Studio. 3+2+2+3
3. (a) Explain widget tree with an example.
- (b) What is the pubspec.yaml file, and what role does it serve in a flutter project?
- (c) What are ternary operators? Explain with example with respect to Dart Programming. 4+2+4
4. (a) Explain the flutter architecture with appropriate illustrations.
- (b) What are Runes? Explain the use of it with suitable examples.
- (c) Define State management in Flutter. 5+3+2

Please Turn Over

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5. (a) What do you mean by Native and Cross platform Development?
(b) What are some of the benefits of using Dart?
(c) Why is Commenting the codes necessary? What are the various ways of commenting applicable in Dart?
(d) Explain while-break with suitable example with respect to Dart Programming. 2+3+2+3
6. (a) What is JSON? Explain the overall structure of JSON with suitable example.
(b) What is Firebase? How is Firebase authentication carried out? Briefly explain with suitable diagram.
(c) What is Cloud Firestore? Highlight the Key features of Cloud Firestore. 3+4+3
7. (a) State the differences between Material Design vs. Cupertino Widgets.
(b) What is the purpose of container? Explain with appropriate illustrations.
(c) What is the purpose of Riverpod in flutter? Explain its benefits. 3+(2+1)+(2+2)
8. Write short notes on (*any two*) : 5×2
- (a) Navigator
 - (b) Stateless widget life cycle
 - (c) Hero animation
 - (d) Maps in Dart Programming.
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